

Community-Based Early Warning System for Water-Borne Diseases

Detailed Project Briefing & Comprehensive Evaluation Guide for External Stakeholders

1. Project Highlights

Real-Time Cluster Warning Detection: One of the most critical parts of this system is its ability to automatically identify localized outbreak clusters without waiting for manual human analysis. When a citizen submits a symptom report, the backend runs a spatial audit over a rolling 7-day window. If the system finds that 3 or more symptomatic reports have been submitted within the same village or district, it automatically triggers a High Risk warning. This warning is saved in the database and immediately sent out to ASHA workers and local villagers, telling them to boil their water and check local water sources, which helps stop a potential outbreak right when it begins.

Optimized Machine Learning Risk Assessment: The platform uses a high-performance Gradient Boosting Classifier to calculate the risk level (Low, Medium, or High) of an outbreak. Unlike standard machine learning pipelines that often use lossy data-compression steps like Principal Component Analysis (PCA) which make the final results hard to explain, our model is trained directly on the raw physical parameters of the water (such as pH level, turbidity, and dissolved oxygen) paired with reported symptoms. This directly improved the model's high-risk classification F1-score to 0.74 and achieved an overall accuracy of 75%, making the predictions highly reliable.

Explainable AI (XAI) Dashboard Cards: To make the system trusted by doctors and health officials, we created an explainable AI visualization interface. Instead of just displaying a raw percentage or risk label, the dashboard renders a detailed card showing the exact inputs fed into the model and a horizontal bar chart displaying the statistical impact of each risk driver (such as a sudden spike in water turbidity or a high rate of diarrhea reports). This allows health officials to look under the hood of the machine learning model and understand exactly why it is predicting high risk for a specific area.

Interactive Outbreak Simulator Sandbox: The official dashboard features an Outbreak Simulator module where health officers can interact with the machine learning model in real-time. By moving simple sliders to adjust water quality parameters (like lowering dissolved oxygen or raising turbidity) and community symptom rates (like raising the percentage of vomiting or diarrhea), the simulator queries the backend Gradient Boosting model and updates the predicted risk level instantly. This is a powerful scenario-planning tool that lets health officers ask 'what-if' questions and prepare response strategies in advance.

2. Project Description

WaterGuard is a specialized digital Early Warning System (EWS) developed to protect communities from the spread of waterborne epidemics. Traditional health monitoring platforms are reactive, logging patient diagnoses days after an outbreak has already peaked and spread through a village. WaterGuard changes this dynamic by creating a proactive, crowdsourced network where citizens, field health workers, and district officers cooperate on a single platform. The system connects clinical health inputs (gastrointestinal symptoms) with environmental parameters (physical water metrics) to identify anomalies early.

By allowing villagers to report stomach sickness and water issues directly from their phones, the platform crowdsources health intelligence. The backend system correlates these reports with the village's water sources (taps, wells, borewells) and uses machine learning to classify the risk. The system provides clear interfaces tailored to each stakeholder: villagers get safe water guidelines and emergency directories, ASHA workers get ground-truth verification checklists and referral forms, and health officers receive district-wide analytics, leaflet mapping hotspots, and direct SMS warning broadcasters to protect the population.

3. System Methodology

The operational methodology of the WaterGuard platform is designed around a closed-loop containment pipeline consisting of six distinct phases. This structured cycle ensures that a report starts as crowdsourced citizen data and ends as an official clinical verification and warning broadcast:

Phase 1 - Localized Input Logging: The cycle starts when a villager logs symptoms (diarrhea, vomiting, fever, abdominal pain, dehydration) or reports a physical water issue (bad smell, discoloration, leak) from their mobile dashboard. The user's coordinates and location are automatically attached to the submission.

Phase 2 - Environmental Correlation: Upon receiving the report, the backend REST API performs a spatial mapping query to identify the user's village. It automatically fetches the latest physical water quality parameters (pH, turbidity, nitrates, sulphates, dissolved oxygen, total suspended solids) recorded for that specific village's infrastructure.

Phase 3 - Machine Learning Assessment: The clinical symptoms and physical water parameters are merged into a single feature vector and passed to the Gradient Boosting model. The model computes the outbreak risk level and the risk probability, showing the top drivers on the user's dashboard.

Phase 4 - Spatial-Temporal Auditing: The database trigger counts symptom reports submitted in the same village and district within the last 7 days. If the count matches or exceeds 3, the system automatically flags a cluster threat and generates a warning notification, applying a 24-hour rate limit to prevent duplicate notification floods.

Phase 5 - Emergency Warning Dispatch: The triggered cluster notification is broadcasted. It appears instantly on the local villagers' dashboards, alerts ASHA workers in the area, and logs simulated SMS warnings to target phone directories in the affected district.

Phase 6 - Field Action & Verification: Local ASHA field workers receive the alerts on their dashboards. They conduct home visits, fill out verification forms with clinical diagnoses, and execute Primary Health Centre (PHC) referrals for high-risk patients. The verified data is fed back into the database to retrain the ML model and improve future predictions, completing the feedback loop.

4. Technologies Used

The platform is built using a modern, lightweight, and robust technology stack that ensures fast processing, smooth rendering, and secure data storage:

Frontend Client: The user interfaces are built using **React (Vite)** for rapid component rendering and hot-reloading. **TailwindCSS** is used to create a clean, responsive layout across desktop and mobile screens. **Recharts** is used to render symptom distribution bar charts and weekly trends. **Leaflet (React-Leaflet)** is integrated to display interactive GIS maps showing outbreak hotspot coordinates and pulsing markers for active warnings.

Backend REST API: The server is built using **Python** and the **Flask** micro-framework, which allows for fast RESTful endpoint mapping. **Flask-SQLAlchemy** is used as the Object-Relational Mapper (ORM) to handle structured queries and database transactions. **Flask-JWT-Extended** handles security, generating secure JSON Web Tokens to restrict dashboard routing based on user roles (Villager, ASHA Worker, Official, Admin).

Machine Learning Pipeline: The predictive risk model is built using **Scikit-Learn's** Gradient Boosting Classifier, which is highly efficient for tabular classification. **Pandas** and **NumPy** are used for preprocessing raw parameters and scaling features. **Joblib** handles the serialization and loading of the trained model (`outbreak_model.pkl`) to execute real-time backend inferences.

Database Management: The system uses structured SQL relational tables (SQLite for local development and MySQL for production). The database schema defines clear relationships between Users, Symptom Reports, Water Sources, Incident Reports, and Notifications (Alerts), enabling fast spatial-temporal join queries.

5. Existing Systems vs. Our Project

To help evaluators understand why this project represents a significant advancement over current protocols, we must examine how existing systems operate and where they fail to prevent epidemics:

Existing Clinical Surveillance: Traditional disease tracking platforms rely entirely on hospital admission logs. When a water source becomes contaminated, villagers get sick, but they do not immediately go to the hospital. There is a lag of 3 to 7 days before patients visit clinics, get diagnosed, and have their records uploaded. By the time health officials notice a spike in cases, the outbreak has already peaked and spread throughout the community. Furthermore, clinical databases are completely separate from municipal water testing records, meaning doctors cannot easily see if patients are drinking contaminated water.

Existing Water Testing Boards: Traditional municipal water boards periodically test local storage tanks and pipelines for chemical and biological parameters (like pH level and turbidity). However, these tests are conducted on a weekly or monthly schedule and logged in static spreadsheets. There is no real-time connection to public health channels. If a water test shows a dangerous bacterial load, there is no automated trigger to warn local clinics or residents. The data sits in silos, useless for immediate epidemic prevention.

WaterGuard EWS (Our Project): WaterGuard bridges the clinical and environmental silos. By collecting symptoms directly from citizens in real-time, the system bypasses hospital admission delays. The backend immediately correlates these symptoms with physical water quality metrics and evaluates risk using machine learning. Rather than relying on manual alerts, our 7-day spatial cluster rule automatically generates containment warnings the moment 3 cases appear in an area. This connects reporting, analysis, verification, and alert broadcasts into a single, unified workflow.

6. Project Modules & Explanations

The platform is divided into six core modules that handle data intake, mathematical modeling, spatial checking, field workers coordination, official analytics, and model maintenance. Here is a detailed breakdown of each module:

Module 1: Crowdsourced Symptom & Contamination Logger (Citizen Interface)

This module serves as the primary data intake channel for the early warning system. It provides villagers with a simple, high-visibility interface to log gastrointestinal health data (diarrhea, vomiting, fever, abdominal pain) and report physical water anomalies (bad smell, leak, discoloration). When a citizen submits a report, the interface automatically captures their location coordinates and details the water source used. This crowdsourcing layer ensures that early symptoms are logged directly from the community, bypassing hospital delay bottlenecks.

Module 2: ML Risk Assessment & Explainability Engine

This module handles risk calculation and transparent explainability. It merges the user's symptoms with the latest local physical water parameters and passes them to the Gradient Boosting classifier. Crucially, the model is trained directly on raw parameters (avoiding lossy compression like PCA). This allows the frontend to render an Explainable AI Card, showing a detailed breakdown of the mathematical weights and feature drivers (such as turbidity spikes) behind the risk score, building trust with clinicians.

Module 3: 7-Day Spatial-Temporal Cluster Audit Rule

This backend rule acts as the automated warning trigger. Immediately after a survey is submitted, the system queries the database to count similar reports submitted in the same village and district within the last 7 days. If the count matches or exceeds 3, the system automatically generates an Outbreak Warning notification, bypassing manual

approval delays to ensure immediate community alerting. A 24-hour rate limit prevents duplicate alert flooding for the same area.

Module 4: ASHA Field Care & Referral Workbook

This module coordinates community field workers. It provides ASHA workers with a tabbed dashboard to manage household care: (a) a Household Checklist to mark family health flags, (b) a Verification Ledger to input ground clinical diagnoses, (c) a High-Risk Referral panel to instantly refer patients to local PHCs, and (d) a Field Visit Log to record visit notes and water storage hygiene checks.

Module 5: Health Official Surveillance & Broadcast Portal

This module provides district health officers with localized analytics tools. It features an interactive Leaflet GIS hotspot map, symptom charts, an emergency SMS warning broadcaster to push warnings to targeted villages or districts, and an interactive outbreak simulator sandbox where officials slide water parameters to preview risk updates.

Module 6: ML Pipeline Model Manager

This module handles retraining and model maintenance. It provides an interface in the Admin dashboard to retrain the Gradient Boosting classifier on new verified clinical diagnoses submitted by ASHA workers. This completes the database feedback loop, allowing the model to continuously learn and improve its predictive accuracy over time.

7. Limitations & Future Enhancements

7.1 Platform Limitations:

1. **Connectivity Dependency:** The early warning system relies on mobile networks or internet access. In extremely remote or forest villages, connectivity lag can delay symptom logging and warning delivery.
2. **Risk of False Logging:** Crowdsourced reporting is susceptible to false or exaggerated logs. To mitigate this, the platform routes reports as 'Pending' until ASHA workers conduct physical home visits to verify symptoms and record clinical diagnoses.
3. **Water Testing Latency:** Environmental water parameters are often updated manually by labs on a weekly or monthly schedule. This means the machine learning model may correlate real-time symptoms with slightly outdated physical water statistics.

7.2 Future Enhancements:

1. **Real-Time IoT Sensors:** Deploy solar-powered physical sensors (measuring pH, turbidity, and chlorine) directly inside village storage tanks to stream real-time water metrics to the EWS API, eliminating manual data upload delays.
2. **Advanced Spatial Clustering:** Integrate spatial clustering algorithms (like DBSCAN) to mathematically model outbreak vectors across administrative borders instead of using static database limits.
3. **Multi-Lingual Voice Reporting:** Deploy an Interactive Voice Response (IVR) phone line, allowing rural citizens without smartphones to report symptoms in their native language (e.g. Telugu) using simple voice prompts.
4. **National Surveillance Sync:** Build secure API integrations with India's Integrated Disease Surveillance Program (IDSP) to synchronize telemetry and coordinate response measures automatically.

8. Conclusion

The WaterGuard platform successfully demonstrates that bridging the gap between crowdsourced clinical health logs and environmental water parameters yields an effective outbreak early warning platform. By replacing slow, manual workflows with automated cluster checks, ML risk assessments, explainable AI cards, and ground-truth verification loops, the system offers an actionable, transparent tool to identify and contain waterborne epidemics before they become serious.